Radiation Survey of Teletherapy and Brachytherapy Units

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1 Objective

- Radiation leakage measurement of telecobalt and brachytherapy units in source-off condition.
- Radiation protection survey of the Teletherapy and Brachytherapy installation.

2 Apparatus

- Radiation Survey Monitor
- Pocket Dosimeter

Radiation Survey Monitor: A radiation survey monitor is a portable device used to measure and detect ionizing radiation levels in various environments. It typically consists of a Geiger-Müller tube or scintillation detector that can detect different types of radiation, such as alpha, beta, gamma, and X-rays. The monitor provides real-time readings of radiation intensity, allowing users to assess potential radiation hazards and ensure safety in areas where radioactive materials are present.

1. **Pressurized Ionization Chamber Survey Meter:** This type of survey meter uses a pressurized ionization chamber to detect and measure ionizing radiation. It is particularly effective for measuring high levels of gamma and X-ray radiation. The pressurized chamber increases the sensitivity of the device, allowing for accurate readings even in low radiation environments.

   Fluke 451P: The Fluke 451P is a specific model of a pressurized ionization chamber survey meter known for its precision and reliability. It is widely used in medical, industrial, and environmental applications to monitor radiation levels and ensure compliance with safety standards. **Key features** of the Fluke 451P include:

   - High µR sensitivity measurement of rate and dose simultaneously
   - Records peak rate using "Freeze Mode"
   - Auto-ranging and auto-zeroing
   - Ergonomic, anti-fatigue handle with replaceable grip, wrist strap and tripod mount
   - Available with dose equivalent energy response (SI units)
   - Programmable flashing LCD display

2. **Geiger-Müller (GM) Survey Meter:** The GM survey meter utilizes a Geiger-Müller tube to detect ionizing radiation. It is widely used for its ability to detect a wide range of radiation types, including alpha, beta, and gamma radiation. The GM survey meter provides audible and visual indications of radiation levels, making it easy to use in various settings.

   Nucleonix RM701N: The Nucleonix RM701N is a specific model of a Geiger-Müller survey meter known for its versatility and ease of use. It is commonly employed in radiation safety, environmental monitoring, and emergency response situations. **Key features** of the Nucleonix RM701N include:

   - Microcontroller based design.
   - Compact, elegant and light weight
   - Uses energy compensated GM Counter.
   - Accuracy+/- 15% with Cs-137.
   - Measures Dose in mR/hr or µSv/hr.
   - Works on 6V (1.5X4V) dry cell batteries.
   - IP rating is better than IP56.
   - Response Time: < 5 seconds.

Pocket Dosimeter: A pocket dosimeter is a small, portable device used to measure an individual's exposure to ionizing radiation over a period of time. It is typically worn on the person, such as on a lapel or pocket, and provides a direct reading of the accumulated radiation dose. Pocket dosimeters are commonly used by radiation workers to monitor their exposure and ensure it remains within safe limits.

Mini Electronic Pocket Gamma Dosimeter: The Digital Dosimeter MPD 1501 is a reliable, lightweight, and convenient personal electronic dosimeter designed for the accurate monitoring of individual X and Gamma radiation exposure.

Features:

- Based on semiconductor diode detector
- Upward replacement for quartz fiber ion chamber dosimeter
- Convenient PEN-type design
- Light weight < 60gm
- Excellent immunity to EMI upto 100V/cm, ERTL certified
- unaffected by Mobile phones in close proximity
- Over 2500 hrs of continuous operation without battery change

Radiation detected	: X and γ radiation
Display Units	: Direct readout of dose equivalent in μSv
Display	: 6-digit LCD with back-lighting (momentary)
Range Dose	: 1μSv to 999999μSv (1 μSv – 1Sv)
Energy response	: ±25% from 50 keV to 3 MeV (ref. 137Cs)
Angular response	: ±15% up to ±60° for 137Cs
Accuracy	: **(^{137}Cs) ±10%**
Dose rate linearity	: ±10% upto 100 mSv/h
	: ±20% up to 1Sv/h
Overload	: **up to 10 Sv**

(a) Flukke 451P

(b) Nucleonix RM701N

(c) Pocket Dosimeter MPD 1501

Figure 1: Various Radiation Survey Instruments

3 Theory

Radiation safety in the handling of radiation sources and radiation-generating equipment, which only covers sealed sources and particle accelerators, is an important aspect for radiation workers and public protection. Radiation surveillance requirements and procedures for medical applications of radiation are specified for ensuring radiation safety of -

- Persons handling radiation-generating equipment and sources for medical applications.
- Patients who undergo medical procedures for their health benefit.

- Persons connected with the patient who is either living with him or assisting him during the medical procedure.
- Public members not related to the medical use of radiation.

In radiation therapy, there are two main types of treatment: teletherapy and brachytherapy. In the case of teletherapy, gamma-emitting sealed radioactive sources (Ex. Co-60) and radiation- generating equipment are used. In the case of brachytherapy sealed Ir-192, a radioactive source is used. When not used, the radioactive sources are kept inside a safe housing in the machine. However, since the intensity of gamma rays follows the exponential law of attenuation, the intensity becomes zero at infinite thickness. Tungsten or lead is mostly used as shielding material for safe housing and is adequate to bring the intensity below the tolerance value. This is confirmed by the radiation protection survey of the equipment, and it ensures the safety of the radiation worker and patient. The rooms designed to install the radiation therapy equipment are also provided with an adequate thickness of the walls, ceiling, roofs, and floors if needed to ensure the safety of the radiation worker, patient, attendant public, etc. These shielding criteria are calculated considering the patient workload, use factor, and the occupancy factor around the installation.

Maximum leakage measurement of teletherapy and brachytherapy unit: While carrying out the survey of radiation equipment and installation, the permissible dose limits for radiation workers and members of the public should be kept in mind. The weekly limit for radiation workers and members of the public is 40mR and 2mR, respectively.

4 Observation

While carrying out the survey of radiation equipment and installation, the permissible dose limits for radiation workers and members of the public should be kept in mind. The weekly limit for radiation workers and members of the public is 40mR and 2mR, respectively.

4.1 Leakage measurement of teletherapy Unit

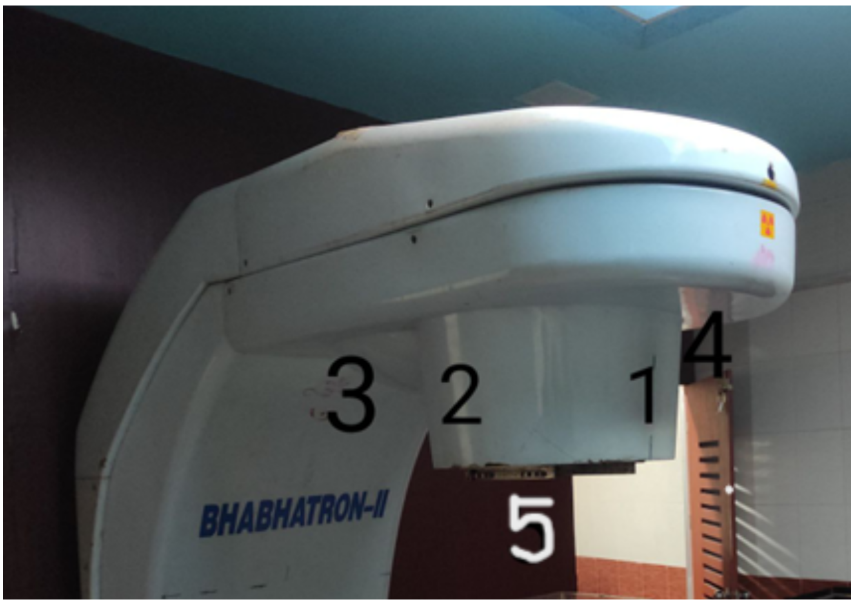

Figure 2: Gantry head of Teletherapy unit

	5cm from the surface of the source housing		
#	Position	Reading ion chamber (mR/hr)	Reading GM survey meter (mR/hr)
1	Front of collimator	1	0.78
2	Left of collimator	1.4	1.09
3	Back of collimator	3.7	3.43
4	Right of collimator	1.52	1.53
5	Underneath of collimeter	-	2.65

	1 m from the surface of the source housing		
#	Position	Reading ion chamber ((μR/hr)	Reading GM survey meter ((μR/hr)
1	Front of collimator	129	0.11
2	Left of collimator	160	1.47
3	Back of collimator	540	0.59
4	Right of collimator	230	1.4
5	Underneath of collimeter	-	0.4

Equipment	Leakage Radiation Dose Rate (μSv/hr)	
Source in the off position loaded with maximum activity	5 cm from the surface of the source housing	1 m from the source
	Measured: 32.4, 30.04	Measured: 4.73, 5.16
	Tol = 200	Tol = 20

Note: measurements shall be averaged over an area not greater than (i) 100 cm2 at 1 m from the source or (ii) 10 cm2 at 5 cm from the surface of the source housing.

4.2 Leakage measurement of Brachytherapy Unit

Brachytherapy unit used Ir-192 source which is having current activity of 8.281 Ci. The source is kept inside the safe housing when not in use. The leakage measurement is carried out using a survey meter around the safe housing of the brachytherapy unit.

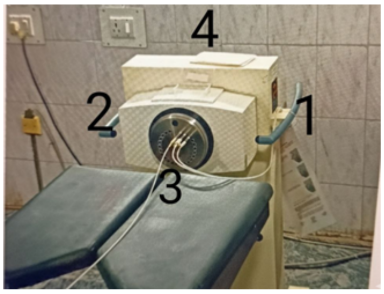

Figure 3: Brachytherapy unit

#	Position	Reading ion chamber (μR/hr)	Reading GM survey meter (μR/hr)
	5cm from the surface of the source housing		
1	Right	624	70
2	Left	124	110
3	Front	171	60
4	Back	430	70

#	Position	Reading ion chamber (μR/hr)	Reading GM survey meter (μR/hr)
	1cm from the surface of the source housing		
1	Right	26	40
2	Left	42	40
3	Front	34	30
4	Back	39	30

Equipment	Leakage Radiation Dose Rate (μSv/hr)	
Remote after-loading system	5 cm from the surface of the source housing	1 m from the source
	Measured: 3.76, 0.96	Measured: 0.367, 0.35
	Tol = 100	Tol = 10

Note: measurements shall be averaged over an area not greater than (i) 100 cm2 at 1 m from the source or (ii) 10 cm2 at 5 cm from the surface of the source housing.

4.3 Survey of teletherapy

The survey of teletherapy unit is carried out using a survey meter around the teletherapy unit in the treatment room. The readings are tabulated below:

Location	Max rad level (μSv/hr)			
	Ion chamber based		GM based	
	0°	270°	0°	270°
A	0.535	0.140	0.175	0.263
B	0.315	0.271	0.263	0.175
C	0.219	0.228	0.087	0.263
D	0.201	0.263	0.263	0.263
E	0.289	0.201	0.263	0.263
F	0.245	0.236	0.263	0.263
G	0.175	0.263	0.263	0.263
H	0.333	0.280	0.263	0.263
I	0.149	0.201	0.263	0.263
J	0.236	0.429	0.263	0.263
M1	0.228	0.175	0.263	0.263
M	0.342	0.210	0.263	0.175
M2	0.228	0.201	0.263	0.175
K	0.885	0.570	0.526	0.350
L	0.385	0.491	0.438	0.701
O	0.385	0.298	0.350	0.526

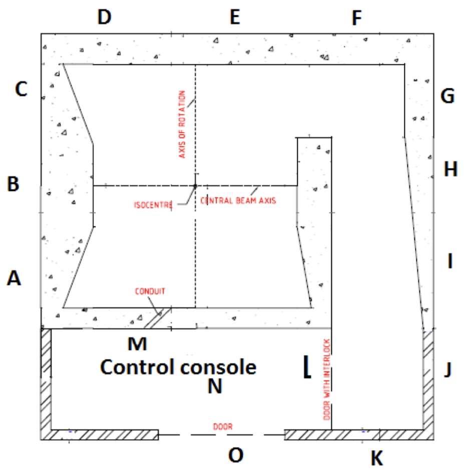

Figure 4: Layout of teletherapy unit in treatment room

4.3.1 Weekly Time Averaged Dose-Equivalent Rate

If R_w is the TADR averaged over a 40 hour week, it can be derived that :

$$R_W = \mathrm{IDR} \times \frac{WU}{\mathrm{DR}_0} \tag{1}$$

R_w is the TADR averaged over one week, in $\mathrm{Sv} \cdot \mathrm{week}^{-1}$;
IDR is instantaneous dose-equivalent rate in $\mathrm{Sv} \cdot \mathrm{h}^{-1}$ when the machine is operating at the dose output rate DR_0;
W is the weekly workload defined at 1 m in Gy; $= 1.5 \times 10^5$ cGy/week
U is the use factor (=1 for secondary barriers or the maze entrance);
DR_0 is the dose output rate at 1 m, in $\mathrm{Gy} \cdot \mathrm{h}^{-1} = 82.38 cGy/min \times (80/100)^2 = 52.7232$.

- For **ion based survey meter**, $R_w = 0.0414$ mSv/week $= 4.719$ mR/week.
- For **GM based survey meter**, $R_w = 0.029$ mSv/week $= 3.306$ mR/week

4.4 Survey of brachytherapy

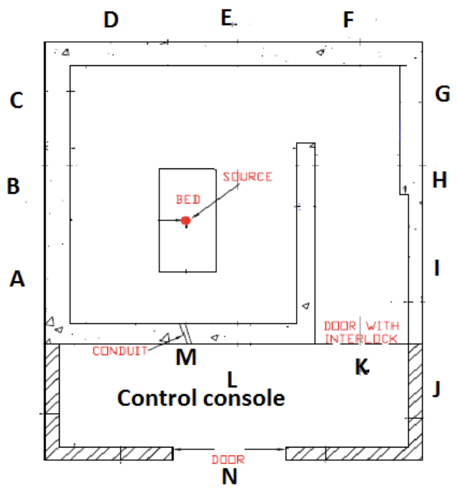

Figure 5: Layout of brachytherapy unit in treatment room

Location	Max rad level (µSv/hr)	
	Ion chamber based	GM based
A	30	50
B	21	60
C	34	30
D	3300	2590
E	2500	1610
F	330	650
G	34	30
H	21	60
I	30	50
J	25	40
M1	340	540
M	42	530
M2	48	340
K	300	500
L	850	650
O	300	-

4.4.1 Weekly Time Averaged Dose-Equivalent Rate

The weekly time averaged dose-equivalent rate is calculated using the formula,

$$R_W = \text{IDR} \times \text{weekly treatment time (4 hr/week)} \times \frac{A_0}{A_t} \tag{2}$$

R_w	is the TADR averaged over one week, in $\text{Sv} \cdot \text{week}^{-1}$;
IDR	is instantaneous dose-equivalent rate in $\text{Sv} \cdot \text{h}^{-1}$ when the machine is operating at the activity A_t;
A_0	is the initial activity of the source, in Ci; = 10 Ci
A_t	is the current activity of the source, in Ci; = 8.281 Ci

- For **ion based survey meter**, $R_w = 3300 \times 10^6$ R/hr $\times$ 4 hr/week $\times \frac{10}{8.2} = 0.01609$ R/week.
- For **GM based survey meter**, $R_w = 2.59 \times 10^{-3}$R/hr $\times$ 4 hr/week $\times \frac{10}{8.2} = 0.01263$ R/week.

5 Conclusion

In conclusion, the radiation survey of teletherapy and brachytherapy units is a critical aspect of ensuring radiation safety for both radiation workers and the public. The implementation of stringent safety protocols, regular monitoring, and adherence to established dose limits are essential to minimize radiation exposure and protect individuals from potential hazards associated with radiation therapy equipment. The survey results indicate that the leakage measurements for both teletherapy and brachytherapy units are within the permissible dose limits, demonstrating the effectiveness of the shielding and safety measures in place. Ongoing vigilance and periodic surveys are recommended to maintain a safe environment in radiation therapy facilities.